

Flexibility in Power System: Market Design Matters

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Abstract

The growing share of renewable energy requires sufficient investment in power system flexibility. In this paper, we frame a three-stage peak-load pricing model consisting of investment, commitment, and real-time operation, considering that electricity generation is costly to adjust on short notice. Adapting the idea of real options theory that waiting is valuable, flexible firms avoid producing in the low-demand state and earn a premium to recoup investment costs. The results demonstrate the importance of increasing time granularity in electricity markets with efficient state-contingent prices.

On top of that, this paper discusses the efficiency of alternative market designs in the investment of flexible assets. In the absence of an efficient real-time market, the forward price results in a distortion in technology investment. This distortion, in theory, can be corrected by an options market, while any centralized auction fails to achieve the optimum. Finally, this work briefly illustrates the effect of demand flexibility, showing that an increase in demand response does not always reduce the reliance on production flexibility if rationing is random.

Keywords: Flexibility, Real-Time Prices, Reserves, Uncertainty, Electricity, Market Design

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1 Introduction

This paper analyzes the value of flexibility and how market design affects investment decisions in flexible assets when quick adjustment is expensive. This question is pertinent to electricity markets, where electricity demand and supply are not perfectly predictable yet must be continuously balanced. With an ambitious investment in global intermittent renewable energy sources (IRES) over the next three decades,¹ The International Energy Agency (hereafter: IEA) predicts a fourfold increase in the demand for flexibility.²

In engineering practice, flexibility is characterized by physical parameters such as ramp rates, start-up times, and minimum load levels. These constraints determine how quickly and by how much a generator can adjust its output. However, adjusting output is not costless. Beyond the marginal production costs, generators incur additional adjustment costs when changing output levels. These adjustment costs may reflect wear and tear, thermal stress, and fuel inefficiency. Furthermore, investments in flexible units incur capital costs,³ and the generators incur commitment cost, such as the standby cost to provide flexibility. All these costs must be compensated to incentivize sufficient flexibility, the key to which is to provide a correct price signal. Electricity reserve markets are designed to guarantee short-term operational flexibility that would otherwise lead to a blackout (Bushnell & Oren, 1994; Cramton, 2017; Wilson, 2002). The consensus is that reserves should be reimbursed for the opportunity cost of not serving in the parallel energy market, plus the additional adjustment and commitment costs. However, cost recovery in the long term is not well addressed.

This paper aims to fill this gap by building a stylized theoretical model to understand the effects of electricity market design on both short-term flexibility operation and long-term investment. Specifically, we address two main research questions: (1) What is the efficient pricing and investment level of flexibility? (2)

¹According to the report "Net Zero by 2050", renewables are expected to generate 88% of global electricity in 2050. Solar PV and wind make up nearly 70% of the total share. As a reference, 29% of electricity is supplied by renewable sources in 2020, and about two-thirds is from hydro-power.

²Flexibility requires more than double by 2030 and 7 times by 2050 in the EU, according to the European Commission. See https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/future-eu-power-systems-renewables-integration-require-7-times-larger-flexibility-2023-06-26_en.

³Even though flexible technologies can have lower capital costs than inflexible ones, improving flexibility for a given technology can incur additional investment costs. For example, solar panels with batteries are flexible but more costly than standard panels without batteries. Coal-fired power plants can improve their operational flexibility by retrofitting with a steam turbine and thermal energy storage.

Which market designs can achieve this efficiency?

We extend the traditional two-stage peak-load pricing model to a three-stage game that introduces commitment between the investment stage and real-time operation. The difference between committed and realized production represents the adjustment component, which is the key setup in this paper. The first main result is that assets that perform output adjustments and thereby provide flexibility should earn a *flexibility premium*, which is called value of waiting in real options theory (Schwartz & Trigeorgis, 2004; Trigeorgis, 1996), defined as the difference between net real options value and net present value. This premium is endogenously reflected in a competitive real-time price and is necessary to recoup the investment costs of flexible technologies. We also find that if rationing is random over consumers, the increase in demand response does not always reduce the value of the generator's flexibility.

Second, we show that a reserve market in which the system operator (SO) provides a menu of option contracts can also achieve the optimum in the absence of real-time pricing.⁴ However, to our best knowledge, the flexibility premium has neither been explicitly recognized in the literature nor systematically incorporated into the compensation mechanisms of previous or existing reserve markets, such as a forward integrated auction of energy (committed electricity production) and reserves (Ehsani et al., 2009; Oren & Sioshans, 2005), a separate reserve market with pay-as-bid pricing (Bushnell & Oren, 1994), or a separate reserve market on the basis of pay-as-clear (Chao & Wilson, 2002). This finding implies that current reserve markets fail to incentivize sufficient investment in flexibility.

This paper relates to three strands of literature. First, it contributes to the peak-load pricing model that is developed to determine efficient pricing and investment in power markets. Modeling periodic and deterministic demand supplied by a single technology, Boiteux (1960)⁵ shows the off-peak price is equal to marginal production cost and consumers during peak hours pay for both production and capacity costs. Crew and Kleindorfer (1976) extend this model to multiple technologies, and the new insight is that both off-peak and peak consumers should pay for capacity on top of production costs. Joskow and Tirole (2007) follow this setup and extend it to a continuum of technologies. It is also recognized that demand is not only periodic, but also uncertain (Brown & Johnson, 1969; Carlton, 1977; Crew & Kleindorfer, 1976; Visscher, 1973). The problem becomes choosing

⁴In the context of this paper, a reserve market plays the role of an option market.

⁵This article was translated by H. W. Izzard from an article in French which appeared in the *Revue générale de l'électricité* in August, 1949.

capacity and price before demand is realized. The results highly depend on the model setup. Visscher (1973) shows that with random rationing, optimal pricing and investment would result in a price lower than long-run marginal cost (hence, allowance is needed to guarantee the zero-profit condition), but capacity is the same as efficient rationing. Carlton (1977) proves that pricing depends on the way uncertainty enters the demand function. If uncertainty enters the demand curve in an additive term, price is lower than long-run marginal cost, but the conclusion is reversed with multiplicative demand uncertainty.

The theoretical progress has stopped since then. With uncertainty, the equilibrium mentioned above is sub-optimal since it lacks state-contingent prices and output. Two points are missing in the literature. First, it assumes all technologies are flexible, while in reality, real-time adjustment is rarely cost-free, and ignoring this cost would lead to inefficient investment. Second, the literature fails to reflect the value of flexibility for flexible assets in prices. This paper extends the framework by including flexibility commitment and adjustment. We model both periodic and uncertain demand net of renewables, but the focus of this paper is efficient investment in flexibility rather than investment at peak time.⁶ van der Weijde and Hobbs (2012) quantify the multi-stage settlement and validate uncertainty in transmission planning. The model in Anupindi and Jiang (2008) has a similar timing as our work, but they concentrate on the strategic equilibria in a duopoly competition where firms are both flexible or inflexible. Nevertheless, our model proposes a monopolistically competitive scenario and reveals the impacts of market organization.

This paper also connects with the long-standing literature on adjustment costs, a concept that is widely used in analyzing stocks Hay (1970), capital investment (Gould, 1968; Lucas, 1967; Schramm, 1970), and labor demand (Jaramillo et al., 1993). Energy economists consider inflexibility in the real-time power market by introducing a steeper supply curve in real-time compared to a forward market. Hortaçsu and Puller (2008) show that supply curves of small generators are rather inelastic in the real-time market. Ito and Reguant (2016) find evidence that for fringe suppliers, adjustment in intra-day markets costs 5 to 10 times more than forward scheduling. We build a theoretical framework to incorporate those observations, and we believe this is necessary when more uncertainty is introduced in the energy transition.

⁶Peak-load technologies are often used to provide flexibility, but this does not mean they are equivalent. Peak-load technologies increase supply for anticipated peak demand, while flexibility expands or curtails output to sudden demand and supply change.

Finally, we contribute to the literature on electricity reserve markets by examining their long-term efficiency. The existing work focuses on auction design that aims for short-term efficiency, addressing asymmetric information in commitment and operational costs. Chao and Wilson (2002) propose that capacity and production should be bid separately: Strategic bidding uses only the capacity component, and energy supplies are paid the spot price. A nonlinear scoring rule with discriminatory pricing also works if generators agree with the system operator on the probability distribution of energy calls (Bushnell & Oren, 1994). Oren and Sioshans (2005) propose an integrated market for both energy and reserves in which the activated reserves are paid the same price as energy, and reserves not activated receive a capacity payment equal to the difference between the clearing price and their own bid.

As for the role of reserves, this paper argues that they provide a way to price flexibility in the absence of a real-time market. Joskow and Tirole (2007) consider operating reserves a public good, so they should be procured centrally to prevent a full system breakdown. In their model, the optimal dispatched load is known, but a fraction of capacity may unexpectedly fail in real time. Hence, it is necessary to provide reserves to avoid system collapse. On top of that, reserves are also viewed as a financial hedge to deal with spot price uncertainty.⁷ In two related papers (Anderson et al., 2017; Wu & Kleindorfer, 2005), reserves are used as an option contract that strategically substitutes for energy in the real-time market. Market equilibrium determines the optimal allocation between reserves and energy based on the distribution of real-time prices.

The rest of the paper is organized as follows. Section 2 describes the three-stage peak-load pricing model and presents the socially optimal investment and operation. Section 3 illustrates the equilibrium of different market designs, and compares with the social optimum. Section 4 concludes.

2 Model

In this section, we present the model setup and the social-optimal equilibrium, which describes equilibrium prices and quantity decisions. We then formally define and explain “flexibility premium”, which is central to this paper.

We develop a three-stage model of electricity generation under uncertainty. In the first stage (investment), the generators choose how much capacity to install from a range of technologies. In the second stage (unit commitment), they commit

⁷For instance: the Short Term Operating Reserve (STOR) in the UK.

to production quantities and reserve levels, i.e., the amount of operational flexibility they schedule, based on partially revealed information about future demand. In the third stage (real-time operation), the actual demand is realized. The key innovation relative to the two-stage model is that generators adjust their output within the limits of their previous commitments. Generation technologies vary in investment cost, marginal production cost, and flexibility costs associated with scheduling reserves and adjusting production in real time.

Electricity demand (net of renewable) is uncertain and changes between stages of the model. Before committing to production, generators receive some information about future conditions, such as time of day, demand, and weather forecasts, but do not observe the exact demand realization. In the unit commitment stage, generators manage this uncertainty by scheduling production and upward and downward reserves. These reserves allow them to adjust the output upward or downward once actual demand is realized. Overcommitting production may result in costly downward adjustments, while undercommitting may require activating expensive upward reserves. In the investment stage, generators can manage uncertainty by investing in technologies that have lower flexibility costs. However, more flexible technologies may involve higher capital or operational costs.

On the demand side, there are two categories of representative consumers: price-sensitive and price-insensitive consumers. Price-sensitive consumers observe real-time market conditions and can adjust their level of consumption accordingly. Price-insensitive consumers do not respond to real-time prices, although their consumption may be rationed when market conditions require it. Price-sensitive consumers buy directly in the wholesale market. Price-insensitive consumers are served by competitive retailers. Retailers do not incur costs for retail activities. They buy electricity in the wholesale market and sell it to consumers under two-part tariffs, corresponding to a marginal energy fee and a lump sum fee.

In our model, inflexibility arises from both the supply side and the demand side. Supply inflexibility reflects the cost of adjusting output on short notice. On the demand side, price-insensitive consumers do not adjust consumption in response to real-time prices. Their demand is perfectly inelastic, unless consumers are rationed.

We assume that all parties in the model are risk-neutral. Generators, retailers, and consumers maximize expected profit and expected net utility. Hence, there is no need for contracts to reallocate risks across agents. Furthermore, we consider a monopolistic competition framework in which all participants are price takers and no individual firm can affect market prices.

2.1 Timing and uncertainty

We now introduce more formally the timing and uncertainty structure of the model. See Figure 1. We represent the gradual resolution of uncertainty by a two-dimensional vector $\xi = (\omega, \varepsilon)$ which describes a possible information trajectory. The interim state ω represents the information that is available at the unit-commitment stage ($T = 2$). The variable ε denotes the final realization of demand at the real-time stage ($T = 3$). The joint density function of information trajectories is given by $h(\xi)$, defined as

$$h(\xi) = g(\omega) \cdot \phi(\varepsilon|\omega), \quad (1)$$

where $g(\omega)$ is the probability density of interim states, and $\phi(\varepsilon|\omega)$ the conditional distribution of the final demand realization ε given ω .

When making investment decisions, firms maximize expected profit over all possible information trajectories, as represented by $h(\xi)$. This is analogous to real option valuation methods in finance, where Monte Carlo simulations are based on simulated state trajectories. The value of flexibility at the interim state ω depends on the dispersion in $\phi(\varepsilon|\omega)$. Commitment decisions are made conditional on the interim information ω , while real-time actions depend on the full information trajectory ξ .

In our model, the interim state ω captures both predictable variation, such as hour-of-day and day-of-week effects, as well as forecasted information about demand, including weather conditions.

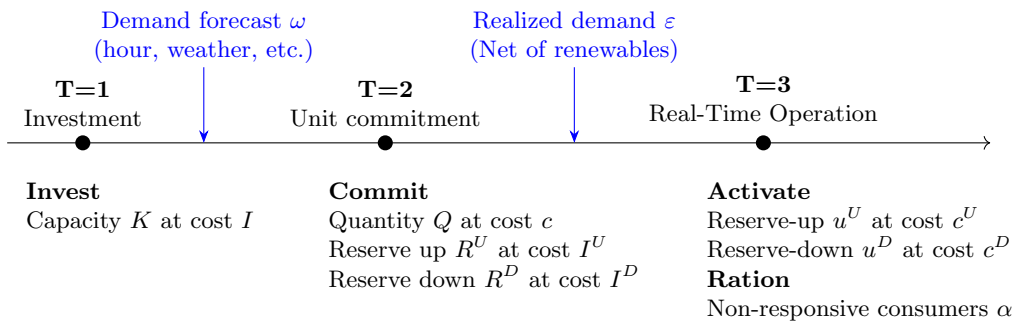


Figure 1: Timing of Decision and Information Revelation

2.2 Supply side

To illustrate the flexibility characteristics of generating technologies in our model, we first introduce a generator with a single technology. We then extend to mul-

multiple technologies. We can describe the generators' actions in the three stages as indicated also in figure 1:

Stage 1 (Investment). The generator chooses the capacity level K , which determines the maximal production capacity in later stages.

Stage 2 (Unit Commitment). Based on forecast information ω , the generator commits to a production level Q_ω , upward reserves R_ω^U , and downward reserves R_ω^D , incurring corresponding commitment costs. The sum of scheduled production and upward reserves needs to be less than installed capacity, and the amount of downward reserves must be less than committed production capacity:

$$R_\omega^D \leq Q_\omega \leq K - R_\omega^U. \quad (2)$$

Stage 3 (Real-Time Operation). Once uncertainty is resolved, a fraction u_ξ^U of the upward reserves and a fraction u_ξ^D of the downward reserves are activated, resulting in additional operating costs. The real-time production q_ξ is the scheduled production level Q_ω plus the fraction u_ξ^D of upward reserves R_ω^U minus the fraction u_ξ^D of downward reserves R_ω^D :

$$q_\xi = Q_\omega + u_\xi^U R_\omega^U - u_\xi^D R_\omega^D \quad \text{with} \quad u_\xi^U, u_\xi^D \in [0, 1]. \quad (3)$$

The total cost incurred in uncertainty trajectory $\xi = (\omega, \varepsilon)$ is:

$$C_\xi = IK + cQ_\omega + (I^U + u_\xi^U c^U)R_\omega^U + (I^D - u_\xi^D c^D)R_\omega^D. \quad (4)$$

This equation summarizes the costs over all three stages. The first term represents the investment cost IK for installing capacity K in stage 1. The second term cQ_ω is the cost for scheduling production Q_ω in stage 2. The third and fourth terms represent the costs associated with scheduling and activating upward and downward reserves. The commitment costs in stage 2 for upward and downward reserves are $I^U R_\omega^U$ and $I^D R_\omega^D$ respectively. In stage 3, for activating upward reserves, the generator pays a per unit cost of c^U , while it recovers c^D per unit of output reduction. Hence, the minus sign for the downward reserves.

Together, equations 2-4 describe the generator's decision problem in managing flexibility: the sequence of actions and the association cost structure. We formally define a generation technology $\theta \in \Theta$ as a set of six cost parameters that represent investment, flexibility, and production characteristics. These parameters are

summarized in Table 1. By varying cost parameters, the model can represent fully flexible, partially flexible, and fully inflexible technologies as special cases.

Table 1: Cost Parameters Defining a Technology θ

Symbol	Description
c_θ	Marginal cost of scheduled production
c_θ^U	Marginal cost of upward adjustment
c_θ^D	Marginal saving from downward adjustment
I_θ	Investment cost per unit of capacity
I_θ^U	Commitment cost per unit of upward reserves
I_θ^D	Commitment cost per unit of downward reserves

To avoid corner solutions and non-trivial commitment decisions, we impose the following assumptions.

Assumption 1 (Cost Parameter Structure). *Each technology $\theta \in \Theta$ is defined by six non-negative cost parameters $I_\theta, c_\theta, I_\theta^U, I_\theta^D, c_\theta^U, c_\theta^D \geq 0$, satisfying the following conditions:*

1. **Marginal cost ordering:**

$$c_\theta^D \leq c_\theta \leq c_\theta^U.$$

It is more efficient to schedule production early, and adjusting production in real time comes at a cost. Downward adjustments allow partial recovery of production costs, but not fully, while upward adjustments are more expensive than scheduled production.

2. **Reserve trade-off condition:**

$$c_\theta - c_\theta^U \leq I_\theta^U - I_\theta^D \leq c_\theta - c_\theta^D. \quad (5)$$

This ensures that both upward and downward reserves can be optimal depending on the probability distribution of demand.

The last condition can be understood by a simple example. Consider a generator that provides flexibility to satisfy random demand: low demand Q_L (with probability ϕ_L) and high demand Q_H (with probability $1 - \phi_L$). It can provide flexibility in two ways: (i) By using an **upward reserve strategy** and scheduling

low output Q_L , and upward reserves $R^U = \Delta Q = Q_H - Q_L$ incurring expected costs:

$$cQ_L + (I^U + c^U(1 - \phi_L))\Delta Q.$$

(ii) By using a **downward reserve strategy** and scheduling high output Q_H and downward reserves $R^D = \Delta Q$, incurring cost:⁸

$$cQ_H + (I^D - c^D\phi_L)\Delta Q$$

The condition in equation (5) ensures that the upward reserves are preferred when low demand is likely ($\phi_L \approx 1$), and downward reserves are preferred when high demand is likely ($\phi_L \approx 0$). Hence, both forms of flexibility can be optimal, depending on the shape of the demand distribution.

Production decisions in stage 3 can be represented by a real-time supply curve as in Figure 2. We illustrate this for three representative technology types:

Panel (2a) represents the case where a technology is **perfectly flexible**, which corresponds to zero commitment costs $I^U = I^D = 0$ and real-time adjustment costs equal to the scheduled marginal production costs $c^U = c^D = c$. Since flexibility is free, the commitment stage becomes irrelevant, and the model reduces to a standard two-stage peak-load pricing model with investments and real-time operation. Generation firm can choose any production quantity q_ξ up to installed capacity K , depending on the realized state.

Panel (2c) shows the case when production technology is **non-responsive**, in which case real-time adjustment is impossible. This arises, for instance, for very high commitment costs for reserves $I^U = I^D = \infty$. This implies that the real-time production q_ξ must be equal to the pre-committed level $q_\xi = Q_\omega$. Absent flexibility, only the information ω available at stage 2 can be used to plan output.

Panel (2b) shows the case for a **limited flexible technology**. This case represents the general setting in our model. The firm commits to a quantity Q_ω , upward and downward reserve R_ω^U and R_ω^D . Real-time production $q_{\theta,\xi}$ in stage 3 must lie in the interval:

$$q_{\theta,\xi} \in [Q_\omega - R_\omega^D, Q_\omega + R_\omega^U].$$

Real-time flexibility exists but is costly: adjustments require prior commitment

⁸The commitment cost of downward reserves is often assumed to be zero ($I^D = 0$) as in Chao and Wilson (2002). We prefer a more general formulation, as it could account for richer interpretations. For instance, a positive I^D may reflect staffing costs to prepare for a production shutdown in response to real-time information.

and incur non-zero marginal costs.

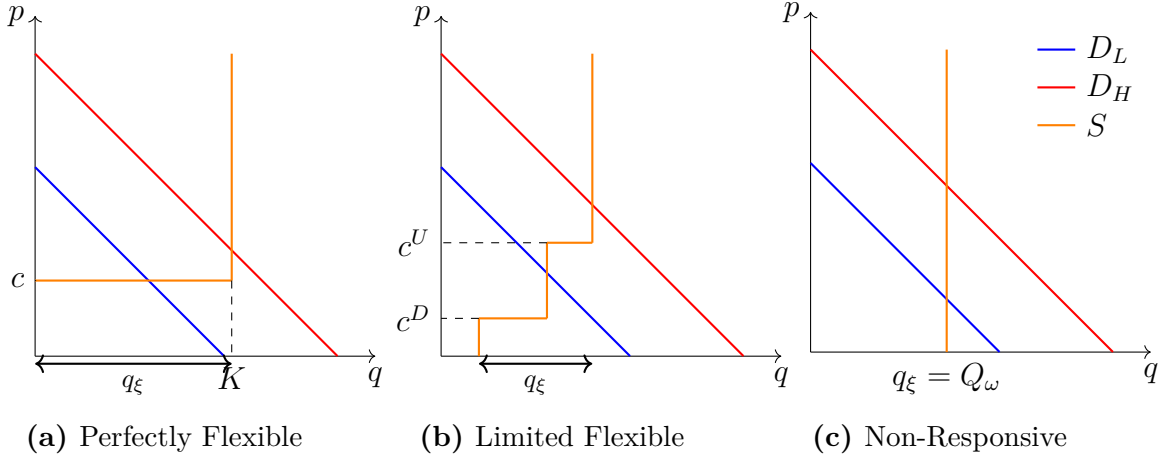


Figure 2: Real-time Supply Functions of Three Technologies

We now extend the model to the set Θ of generation technologies. Each technology $\theta \in \Theta$ is characterized by its own cost parameters and decision variables. The total cost in state ξ is given by:

$$C_{\Theta, \xi} = \sum_{\theta \in \Theta} C_{\theta, \xi},$$

where $C_{\theta, \xi}$ is defined as in Equation 4 for each technology. We also define the aggregate real-time generation decisions across all technologies:

$$q_{\Theta, \xi} = \sum_{\theta \in \Theta} q_{\theta, \xi}.$$

2.3 Demand side

We model consumer flexibility and rationing following the framework of Joskow and Tirole (2007). Before distinguishing between consumer types and their information sets, we first introduce the general setup and notation. Let $D(p)$ denote the demand function, which gives the quantity consumers wish to consume at a price p , and let $S(p)$ be the associated *gross* consumer surplus. The surplus function is decreasing and concave in price.⁹

In some states, demand may exceed available supply and must be rationed. Let α denote the fraction of demand served (i.e., not rationed). The level of demand

⁹The function $S(p)$ is the area under the demand curve $D(p)$. The two functions are related through the identity $S'(p) = pD'(p)$ and are both zero $S(\bar{p}) = D(\bar{p}) = 0$ at the reservation price $\bar{p}$.

served is represented by the calligraphic letter $\mathcal{D}$ and given by:¹⁰

$$\mathcal{D}(p, \alpha) = \alpha D(p).$$

Let the function $\mathcal{S}(p, \alpha)$ denote the gross consumer surplus when the price is p and a fraction α of demand is served. The shape of this function depends on the rationing technology. Under random rationing, we have:

$$\mathcal{S}^{\text{Rand}}(p, \alpha) = \alpha S(p). \quad (6)$$

Smart meters may make rationing more efficient by rationing lower-valued consumption first. Under efficient rationing, the expression becomes:

$$\mathcal{S}^{\text{Eff}}(p, \alpha) = S(p^*) \quad \text{with} \quad D(p^*) = \alpha D(p). \quad (7)$$

Following Borenstein and Holland (2005), consumers are divided into two categories: a fraction σ is price-insensitive, and the remaining fraction $(1 - \sigma)$ is price-sensitive. The parameter σ is taken as exogenous. Price-sensitive consumers adjust their consumption in response to real-time information ξ , but price-insensitive consumers can only adjust consumption based on the price information available in the forward state ω .

The served demand and gross surplus are defined as follows. For price-insensitive consumers:

$$\mathcal{D}_\xi(p_\omega, \alpha_\xi), \quad \mathcal{S}_\xi(p_\omega, \alpha_\xi), \quad (8)$$

and for price-sensitive consumers :

$$\hat{\mathcal{D}}_\xi(p_\xi, \hat{\alpha}_\xi), \quad \hat{\mathcal{S}}_\xi(p_\xi, \hat{\alpha}_\xi), \quad (9)$$

where α_ξ and $\hat{\alpha}_\xi$ denote the fraction of demand served, p_ω is the price paid by price-insensitive consumers (set in advance), and p_ξ is the real-time price faced by price-sensitive consumers.

We defined the *value of lost load* (VOLL) as the marginal social value of restoring curtailed demand:

$$VOLL_\xi = \frac{\partial \mathcal{S}_\xi}{\partial \alpha_\xi} / \frac{\partial \mathcal{D}_\xi}{\partial \alpha_\xi}. \quad (10)$$

¹⁰Joskow and Tirole (2007) argue that this relation does not hold, $\mathcal{D} \neq \alpha D$, if consumers adjust their behavior in response to higher levels of rationing, for instance by avoiding elevators.

2.4 Social optimum

We now describe the social optimum of the three-stage model presented above. The social optimum requires choosing investment capacity K_θ , commitments for energy $Q_{\theta,\omega}$, upward reserves $R_{\theta,\omega}^U$, downward reserves $R_{\theta,\omega}^D$, the reserves' utilization rates $u_{\theta,\xi}^U$, $u_{\theta,\xi}^D$, the price p_ω , p_ξ for price-insensitive and price-sensitive consumers respectively, and their rationing rates α_ξ , $\hat{\alpha}_\xi$, to maximize the social welfare function:

$$\begin{aligned}
\max \quad & \mathbb{E}_\xi \left\{ \sigma \mathcal{S}_\xi(p_\omega, \alpha_\xi) + (1 - \sigma) \hat{\mathcal{S}}_\xi(p_\xi, \hat{\alpha}_\xi) - C_{\Theta,\xi} \right\}, \\
\text{s.t.} \quad & \sigma \mathcal{D}_\xi(p_\omega, \alpha_\xi) + (1 - \sigma) \hat{\mathcal{D}}_\xi(p_\xi, \hat{\alpha}_\xi) \leq q_{\Theta,\xi} & [\eta_\xi h_\xi] \\
& Q_{\theta,\omega} + R_{\theta,\omega}^U \leq K_\theta & [\lambda_{\theta,\omega}] \\
& R_{\theta,\omega}^D \leq Q_{\theta,\omega} & [\mu_{\theta,\omega}] \\
& Q_{\theta,\omega} \geq 0 & [\varphi_{\theta,\omega}^Q] \\
& R_{\theta,\omega}^D \geq 0 & [\varphi_{\theta,\omega}^D] \\
& R_{\theta,\omega}^U \geq 0 & [\varphi_{\theta,\omega}^U] \\
& u_{\theta,\xi}^U, u_{\theta,\xi}^D \in [0, 1].
\end{aligned} \tag{11}$$

Let $\eta_\xi h_\xi$ denote the multiplier of energy balance in real-time state ξ , $\lambda_{\theta,\omega}$ the capacity value of technology θ in state ω , and $\mu_{\theta,\omega}$ the energy value of technology θ in state ω . The last three inequalities are non-negative constraints.

Proposition 1. *Given cost parameters and state distributions $g(\omega), \phi(\varepsilon|\omega)$, the first-order conditions representing the optimum of maximization problem (11) are*

(a) *Price-sensitive consumers:*

No rationing

$$\forall \xi \quad \hat{\alpha}_\xi = 1. \tag{12}$$

Price equal to energy shadow price

$$\forall \xi \quad p_\xi = \eta_\xi. \tag{13}$$

(b) *Price-insensitive consumers:*

Optimal rationing

$$\forall \xi \quad VOLL_\xi = \frac{\partial \mathcal{S}_\xi}{\partial \alpha_\xi} / \frac{\partial \mathcal{D}_\xi}{\alpha_\xi} = p_\xi \quad \text{or} \quad \alpha_\xi \in \{0, 1\}. \quad (14)$$

Retail price p_ω balances deadweight loss

$$\forall \omega \quad \mathbb{E}_{\xi|\omega} \left[\frac{\partial \mathcal{S}_\xi}{\partial p_\omega} - p_\xi \frac{\partial \mathcal{D}_\xi}{\partial p_\omega} \right] = 0. \quad (15)$$

(c) *Efficient activation of reserves.*

$$\forall \xi, \theta \quad u_{\theta,\xi}^D = \begin{cases} 1, & p_\xi < c_\theta^D \\ \in [0, 1], & p_\xi = c_\theta^D \\ 0, & p_\xi > c_\theta^D \end{cases} \quad u_{\theta,\xi}^U = \begin{cases} 1, & p_\xi > c_\theta^U \\ \in [0, 1], & p_\xi = c_\theta^U \\ 0, & p_\xi < c_\theta^U \end{cases} \quad (16)$$

(d) *Commitment.*

Energy commitments are like forward contracts

$$\pi_{\theta,\omega}^Q = \mathbb{E}_{\xi|\omega}[p_\xi] - c_\theta = \lambda_{\theta,\omega} - \mu_{\theta,\omega} - \varphi_{\theta,\omega}^Q. \quad (17)$$

Downward reserves are like put options

$$\pi_{\theta,\omega}^D = \mathbb{E}_{\xi|\omega}[\max\{c_\theta^D - p_\xi, 0\}] - I_\theta^D = \mu_{\theta,\omega} - \varphi_{\theta,\omega}^D. \quad (18)$$

Upward reserves are like call options

$$\pi_{\theta,\omega}^U = \mathbb{E}_{\xi|\omega}[\max\{p_\xi - c_\theta^U, 0\}] - I_\theta^U = \lambda_{\theta,\omega} - \varphi_{\theta,\omega}^U. \quad (19)$$

(e) *Capacity Investment.*

$$\begin{aligned} \mathbb{E}_\omega[\lambda_{\theta,\omega}] &= I_\theta \quad \text{if} \quad K_\theta > 0 \\ &\text{otherwise, } K_\theta = 0. \end{aligned} \quad (20)$$

We interpret the first-order conditions as follows: Proposition 1 (a) indicates that price-sensitive consumers should face the real-time price p_ξ and are never rationed, as in Joskow and Tirole (2007). The intuition is straightforward that price-sensitive consumers can fully adjust their consumption according to state-

contingent prices, and any curtailment is a dead-weight loss.

Proposition 1 (b) gives the optimal conditions for price-insensitive consumers. First, in case of rationing, the optimal rationing rate in state ξ should be such that $VOLL$ in that state $VOLL_\xi$ is equal to the real-time price p_ξ . Note that the real-time price is equal to the shadow price of the production constraint and reflects the social value of energy in state ξ .

The retail price p_ω that consumers pay in state ω is such that the marginal deadweight losses across states ξ are minimized. When there is no rationing, $\alpha_\xi = 1$. With random rationing, the optimal retail price that consumers pay is equal to the expected real-time price plus a correction reflecting the covariance between the real-time price and the normalized demand slope across states ξ :

$$p_\omega = \mathbb{E}_{\xi|\omega}(p_\xi) + \frac{\text{Cov}_{\xi|\omega}[p_\xi, \alpha_\xi \partial D_\xi / \partial p_\omega]}{\mathbb{E}_{\xi|\omega}(\alpha_\xi \partial D_\xi / \partial p_\omega)}.$$

Proposition 1 (c) shows that for scheduled downward reserves, only technologies for which the marginal production cost is larger than the real-time price are activated. And for scheduled upward reserves, only technologies for which the marginal production cost is smaller than the real-time price are activated. Note that in real time, neither capacity investment nor flexibility commitment is reversible, so any associated costs are sunk. The marginal cost only depends on marginal production cost c_θ^U and c_θ^D , respectively.

Proposition 1 (d) gives the necessary first-order conditions in the scheduling stage ($T = 2$). The three expressions on the left-hand side of the equation indicate the values of committing to sell energy (π_ω^Q), downward reserves (π_ω^D), and upward reserves (π_ω^U). Committing to sell energy is like a forward contract, downward reserves are like a put option on the real-time price, and upward reserves are like a call option on the real-time price. Let $\pi^c(X, P_\omega)$ and $\pi^p(X, P_\omega)$ denote the expected return on a call option and a put option with strike price X and option price P_ω . That is

$$\pi_\omega^c(X, P_\omega) = \mathbb{E}_{\xi|\omega}[\max\{p_\xi - X, 0\}] - P_\omega. \quad (21)$$

$$\pi_\omega^p(X, P_\omega) = \mathbb{E}_{\xi|\omega}[\max\{X - p_\xi, 0\}] - P_\omega. \quad (22)$$

then $\pi_\omega^D = \pi^{put}(c^D, I^D)$ and $\pi_\omega^U = \pi^{call}(c^U, I^U)$. However, the commitment decisions are not financial contracts that can be sold independently; instead, they need to be backed up by generation capacity. There are the following inter-linkages: (1) In order to provide downward reserves, a generator needs to be scheduled to

run in the first place. (2) A generation capacity can provide energy or upward reserves, but not both at the same time. Those linkages are described through the Lagrange multipliers on the right side of the equation.

The social planner will use the capacity of technology θ in state ω to maximize its social value $\lambda_{\theta,\omega}$, which is given by:

$$\lambda_{\theta,\omega} = \max \left\{ 0, \pi_{\theta,\omega}^U, \pi_{\theta,\omega}^Q + \max \left\{ \pi_{\theta,\omega}^D, 0 \right\} \right\}. \quad (23)$$

The social planner can decide (1) to keep capacity idle and receive a value 0, (2) commit to upward reserves and make $\pi_{\theta,\omega}^U$, (3a) schedule energy and receive $\pi_{\theta,\omega}^Q$, or (3b) schedule energy and downward reserves $\pi_{\theta,\omega}^Q + \pi_{\theta,\omega}^D$.

Proposition 1(e) is the standard free-entry condition for investment of technology θ . Each investment opportunity earns zero profit. If the expected profit is negative because of a cost disadvantage, there is no investment in this technology.

2.5 Value of flexibility

The trade-off between commitment alternatives depends on the commitment costs, adjustment costs, and state distribution. Lemma 1 derives expressions for the value of generation capacity for two extreme cases: perfectly flexible generation and non-responsive generation, and shows that the value of a limited flexible technology is between those two extremes.

Lemma 1. *For the perfectly flexible and non-responsive technologies, the value of technology θ is given by:*

$$\lambda_{\theta,\omega}^{flex} = \mathbb{E}_{\xi|\omega} \{ \max \{ p_{\xi} - c_{\theta}, 0 \} \} \quad (24)$$

$$\lambda_{\theta,\omega}^{nr} = \max \{ \mathbb{E}_{\xi|\omega} [p_{\xi}] - c_{\theta}, 0 \}. \quad (25)$$

The value of a limited flexible technology is smaller than that of a completely flexible technology and larger than that of a non-responsive technology with the same marginal costs c_{θ} :

$$\lambda_{\theta,\omega}^{flex} \geq \lambda_{\theta,\omega} \geq \lambda_{\theta,\omega}^{nr}.$$

Proof. We need to calculate $\lambda_{\omega,\theta}$ in Equation 23. (i) For the perfectly flexible technology, there is no commitment cost for downward reserves, $I_D = 0$, so it is always optimal to commit to downward reserves as $\pi_{\theta,\omega}^D \geq 0$. The social planner has the choice between committing upward reserves, fully scheduling production, and committing to downward reserves, or leaving capacity idle. Given that $c^D =$

$c^U = c$ and there are no reservation costs, the two reserve strategies give the same profits $\pi_{\theta,\omega}^U = \pi_{\theta,\omega}^Q + \pi_{\theta,\omega}^D$. Those profits are weakly positive, so it is always optimal to provide flexibility. This proves equation 24. (ii). For the non-responsive technology, reserves are prohibitively expensive $\pi_{\theta,\omega}^U < 0$ and $\pi_{\theta,\omega}^D < 0$, so $\lambda_{\theta,\omega}^{nr} = \pi_{\theta,\omega}^Q$, which proves equation 25. $\square$

Comparing the value of a limited flexible technology with that of a non-responsive technology, we can define a flexibility premium $\nu_{\theta,\omega}$.

Definition 1. *The flexibility premium for technology θ in state ω is*

$$\nu_{\theta,\omega} \equiv \lambda_{\theta,\omega} - \lambda_{\theta,\omega}^{nr}.$$

The flexibility premium ν measures the additional value that a flexible asset creates compared to a non-responsive asset with the same marginal cost c_θ . A non-responsive asset must follow the scheduled production and cannot adjust output with the realized demand, while a flexible asset can increase production when the real-time price is high and reduce production when the price is low. This ability to respond to real-time conditions allows the flexible asset to avoid losses in unfavorable states and capture higher-than-average revenue in favorable states. Therefore, the flexible assets have the right, instead of an obligation, to produce or not. The resulting gain from this option value relative to an inflexible asset that can only earn expected revenue is the flexibility premium. By construction, the flexibility premium is non-negative. Lemma 2 shows that the flexibility premium ν in the forward market (commitment stage) can be seen as a combination of put and call options on the real-time price p_ξ .¹¹

Lemma 2. *The flexibility premium is the maximum of two put options when the expected real-time price is larger than c and the maximum of two call options otherwise, but can never become negative:*

$$\nu_{\theta,\omega} = \begin{cases} \max\{0, \pi_\omega^p(c^D, I^D), \pi_\omega^p(c^U, I^U + c^U - c)\} & \text{if } \mathbb{E}_{\xi|\omega}[p_\xi] > c \\ \max\{0, \pi_\omega^c(c^U, I^U), \pi_\omega^c(c^D, I^D + c - c^D)\} & \text{if } \mathbb{E}_{\xi|\omega}[p_\xi] \leq c \end{cases} \quad (26)$$

Example We illustrate the decision procedure and the flexibility premium with a simple numerical example. A firm with a single unit of technology θ needs to commit to energy and reserves in state $\omega \in \{1, 2, 3, 4\}$, given that real-time prices can be either $p_L = 0$ or $p_H = 4$ with probabilities $\phi(L|\omega)$ and $\phi(H|\omega)$.

¹¹See Appendix B for proof.

The technological parameters of technology θ are summarized in Table 2. The optimal commitment strategy is found by calculating π^Q , π^D , and π^U and selecting the option that maximizes the value of capacity using Equation 23. The results are shown in Table 3. In this example, commitment decisions change as the distribution over high and low prices changes. Energy commitment occurs if prices are likely to be high. With more uncertainty, energy commitment and downward reserves are optimal. If high prices are less likely, upward reserves are more attractive. If high prices are very unlikely, it might be better not to commit to the unit at all. If each state ω occurs with probability $1/4$ the total value is $E_\omega(\lambda_\omega) = (2 + 21/20 + 3/16)/4 = 259/320$.

Parameter	Value
Marginal cost	$c = 1$
Upward adjustment cost	$c^U = \frac{5}{4}$
Downward adjustment cost	$c^D = \frac{1}{2}$
Upward commitment cost	$I^U = \frac{1}{2}$
Downward commitment cost	$I^D = \frac{1}{5}$
Firm's capacity	$K = 1$

Table 2: Technological Assumptions

ω	$\phi(L \omega)$	$\phi(H \omega)$	p^F	π^Q	π^D	π^U	λ_ω	Commitment	ν
1	$\frac{1}{4}$	$\frac{3}{4}$	3	2	$-\frac{3}{40}$	$\frac{25}{16}$	2	Q	0
2	$\frac{1}{2}$	$\frac{1}{2}$	2	1	$\frac{1}{20}$	$\frac{7}{8}$	$\frac{21}{20}$	Q and R^D	$\frac{1}{20}$
3	$\frac{3}{4}$	$\frac{1}{4}$	1	0	$\frac{7}{40}$	$\frac{3}{16}$	$\frac{3}{16}$	R^U	$\frac{3}{16}$
4	$\frac{9}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$-\frac{3}{5}$	$\frac{1}{4}$	$-\frac{9}{40}$	0	None	0

Table 3: Profitability and Optimal Strategy under Different States ω .

Demand flexibility To conclude this section, we brief discuss how the proportion of price-insensitive consumers σ affect the flexibility premium. The first-order conditions in Proposition 1(b) are independent of σ . However, the share of consumers affects total demand, which in turn has an impact on cumulative density function of the price distribution $F_\omega(p_\xi)$. First, we show in 3 that a more volatile price distribution would result in larger flexibility premium.

Lemma 3. *If F_{ω_1} second-order stochastically dominates F_{ω_2} , $F_{\omega_1} \succeq_{SSD} F_{\omega_2}$, the flexibility premium in ω_1 is not larger than in state ω_2 :*

$$\nu_{\theta, \omega_1} \leq \nu_{\theta, \omega_2}.$$

Proof. The value of call and put options increases when the tail of the distribution becomes more important or

$$\pi_{\omega_1}^C(X, P) = \int_{\underline{p}}^X (F_{\omega_1}(p) - 1)dp - P \leq \int_{\underline{p}}^X (F_{\omega_2}(p) - 1)dp - P = \pi_{\omega_2}^C(X, P), \quad (27)$$

$$\pi_{\omega_1}^P(X, P) = \int_{\underline{p}}^X F_{\omega_2}(p)dp - P \leq \int_{\underline{p}}^X F_{\omega_1}(p)dp - P = \pi_{\omega_2}^P(X, P). \quad (28)$$

The inequality follows from the definition: distribution $F_{\omega_1}(x)$ has second-order stochastic dominance over $F_{\omega_2}(x)$ if and only if $\int_{\underline{p}}^{p^*} F_{\omega_1}(x)dx \leq \int_{\underline{p}}^{p^*} F_{\omega_2}(x)dx$. This means that each term in the two parts of equation 26 is lower with ω_1 . $\square$

Then, we discuss what implies if there are a larger share of price-sensitive consumers (i.e. σ is smaller). First, if efficient rationing is possible, $VOLL_{\xi} = p_{\xi}$, the demand of price-insensitive consumers does not change when they are rationed or they react to real-time price. However, in the no-rationing case, the total demand increases when price-insensitive consumers become price-sensitive since $p_{\omega} > p_{\xi}$. Hence, the real-time price would increase, and because a first-order stochastic dominance is a sufficient condition of second-order stochastic dominance. The flexibility premium decreases.

When rationing can only be done randomly, denote $\hat{p}_{\xi}$ as the average $VOLL$ and rationing happens if real-time price is higher than $\hat{p}_{\xi}$. Demand and real-time prices increase in the rationing case ($\hat{p}_{\xi} < p_{\xi}$), and in the case where the real-time price is smaller than the marginal price ($p_{\xi} < p_{\omega}$), while it decreases in the case where there is no rationing but the real-time price is larger than marginal price ($p_{\omega} < p_{\xi} < \hat{p}_{\xi}$). Therefore, on average, the change of flexibility value is ambiguous.

The intuition of this result is: when total demand is more elastic, there is less risk of curtailment, so less cost of production commitment (curtailment effect). But this elasticity also implies less demand when the price is moderately high (price effect). Therefore, there are two countervailing effects, and for those technologies with low curtailment cost but high production cost, price effects can dominate, and production commitment becomes less attractive when there are more price-sensitive consumers. Hence, demand flexibility does not necessarily reduce the requirement for production flexibility. I summarize the results in Lemma 4.

Lemma 4. (1) *If rationing can be done efficiently, when the share of price-sensitive consumers increases, production commitment becomes more attractive;*
(2) *If rationing is random, production commitment becomes more attractive in state ω when there is large probability of extreme cases (high demand & low demand), while flexibility is more valuable when moderate cases are more likely to happen.*

3 Effects of market design on flexibility investment

We now compare different electricity market designs with the social-planner benchmark. We first show that a decentralized real-time market can achieve the social optimum equilibrium. Note that because we introduce price-insensitive consumers, the First Welfare Theorem does not automatically apply and requires two conditions. First, there must be competitive retailers that charge a two-part tariff to these price-insensitive consumers and can ration consumers as needed. Second, all price-insensitive consumers must be homogeneous. We then show that, in the absence of a decentralized real-time market, a reserve market combined with a forward market can also achieve the optimum. Finally, we discuss the prevalent reserve market designs and show that none of them provides sufficient flexibility.

3.1 Decentralized real-time market

Definition 2. *Let x_θ be the vector of decision variables of a firm with technology θ : production levels $q_{\theta,\xi}$, unit commitments $Q_{\theta,\omega}$, $R_{\theta,\omega}^U$, $R_{\theta,\omega}^D$, investments K_θ ,*

$$x_\theta = (q_{\theta,\xi}, Q_{\theta,\omega}, R_{\theta,\omega}^U, R_{\theta,\omega}^D, K_\theta).$$

Let y be the vector of decision variables of a retailer selling to homogeneous non-responsive consumers: a fixed fee $\mathcal{A}_\omega$, a retail price for non-responsive consumers p_ω , and the rationing rules α_ξ :

$$y = (\mathcal{A}_\omega, p_\omega, \alpha_\xi).$$

Let z be the vector of price variables: the real-time price p_ξ and the reservation utility of non-responsive consumers U_ω :

$$z = (p_\xi, U_\omega).$$

The vectors x_θ^ , y^* , z^* form a competitive equilibrium if*

- for a given z^* , x_θ^* maximizes the profit of a generator with technology θ ,
- for a given z^* , y^* maximizes the retailers profit,
- in each state ξ all markets clear:

$$q_{\Theta,\xi}^* = \sigma \mathcal{D}_\xi(p_\omega^*, \alpha_\xi^*) + (1 - \sigma) \hat{\mathcal{D}}_\xi(p_\xi^*).$$

- retailers make zero profit,
- the reservation utility of non-responsive consumers U_ω is consistent with the equilibrium contracts.

Proposition 2. *The competitive equilibrium is equivalent to the social optimum.*

Proof. The profit maximization problem of a generator with technology θ is given by:

$$\begin{aligned} x_\theta^* \in \arg \max_{x_\theta} \mathbb{E}_\xi \left[p_\xi q_{\theta,\xi} - I_\theta K_\theta - c_\theta Q_\omega - (I_\theta^U + u_{\theta,\xi}^U c_\theta^U) R_{\theta,\omega}^U - (I_\theta^D - u_{\theta,\xi}^D c_\theta^D) R_{\theta,\omega}^D \right]. \\ \text{s.t.} \quad K_\theta \geq 0 \\ \forall \omega, \quad R_{\theta,\omega}^D \leq Q_{\theta,\omega} \leq K_\theta - R_{\theta,\omega}^U \quad \text{and} \quad Q_{\theta,\omega}, R_{\theta,\omega}^U, R_{\theta,\omega}^D \geq 0 \\ \forall \xi, \quad q_{\theta,\xi} = Q_{\theta,\omega} + u_\xi^U R_\omega^U - u_\xi^D R_\omega^D \quad \text{and} \quad u_\xi^U, u_\xi^D \in [0, 1]. \end{aligned} \tag{29}$$

It is easy to verify that the first order conditions are given by the equation system from (c) to (e) in Proposition 1.

The retailer's profit maximization is given by:

$$\begin{aligned} y^* \in \arg \max_y \mathbb{E}_\xi [\mathcal{A}_\omega + (p_\omega - p_\xi) \mathcal{D}_\xi(p_\omega, \alpha_\xi)]. \\ \text{s.t.} \quad \forall \omega, \quad \mathbb{E}_{\xi|\omega} [\mathcal{S}_\xi(p_\omega, \alpha_\xi) - p_\omega \mathcal{D}_\xi(p_\omega, \alpha_\xi) - \mathcal{A}_\omega] \geq U_\omega. \end{aligned} \tag{30}$$

Let ζ_ω be the multiplier of the constraint. The first order conditions of the retailers are:

$$\begin{aligned} \zeta_\omega = 1 \\ \mathbb{E}_{\xi|\omega} \left\{ \mathcal{D}_\xi + (p_\omega - p_\xi) \frac{\partial \mathcal{D}_\xi}{\partial p_\omega} + \zeta_\omega \left[\frac{\partial \mathcal{S}}{\partial p_\omega} - \mathcal{D}_\xi - p_\omega \frac{\partial \mathcal{D}_\xi}{\partial p_\omega} \right] \right\} = 0 \\ (p_\omega - p_\xi) \frac{\partial \mathcal{D}_\xi}{\partial \alpha_\xi} + \zeta_\omega \left[\frac{\partial \mathcal{S}}{\partial \alpha_\xi} - p_\omega \frac{\partial \mathcal{D}_\xi}{\partial \alpha_\xi} \right] = 0. \end{aligned}$$

Those equations are equivalent to the first-order conditions of the social planner characterized by (b). The zero profit condition determines the fixed part $\mathcal{A}_\omega$:

$$\mathcal{A}_\omega = \mathbb{E}_{\xi|\omega}[(p_\xi - p_\omega)\mathcal{D}_\xi(p_\omega, \alpha_\xi)].$$

This implies that consumers pay a fixed fee $\mathcal{A}_\omega$ if the volume-weighted real-time price p_ξ differs from the contract price p_ω . $\square$

3.2 Reserve Market

We assume the absence of a real-time market and introduce a forward market and a reserve market. At the forward stage ω , generators and retailers trade forward contracts for certain delivery and reserve contracts that take the form of call options on energy with strike price (energy payment) $X \in \mathcal{X}$. The set $\mathcal{X} = \{c_\theta^U, c_\theta^D : \theta \in \Theta\}$ corresponds to the marginal adjustment costs of all technologies. We consider only call options, since put options that represent downward flexibility can be replicated by combining forward and call positions. Forward and option prices (capacity payment), p_ω^F and $\{p_\omega^X\}_{X \in \mathcal{X}}$, are determined endogenously in competitive equilibrium.

There is no real time wholesale market; all adjustments occur through the activation of reserve options by retailers. At the realization stage ξ , retailers balance demand using reserve activation, rationing for non-responsive consumers, and pricing for responsive consumers. Generators supply their contracted forward quantities and any activated reserves.

Definition 3 (Reserve-Market Equilibrium). *A reserve-market equilibrium is a collection*

$$(x_\theta^*, y^*, z^*, e^*),$$

where the price vector z collects all state-contingent variables across forward and real-time states:

$$z = (p_\omega^F, p_\omega^X, p_\omega, U_\omega^{NR}, U_\xi^R),$$

with p_ω^F is the wholesale forward price, p_ω^X the option price with strike price $X \in \mathcal{X}$, p_ω the retail price for non-responsive consumers, U_ω^{NR} and U_ξ^R the reservation utilities of non-responsive and responsive consumers,

where x_θ represents firm θ 's decisions:

$$x_\theta = (K_\theta, Q_{\theta,\omega}, R_{\theta,\omega}^U, R_{\theta,\omega}^D, S_{\theta,\omega}^F, S_{\theta,\omega}^X, u_{\theta,\xi}^U, u_{\theta,\xi}^D)$$

with K_θ the investment levels, $Q_{\theta,\omega}$, $R_{\theta,\omega}^U$, $R_{\theta,\omega}^D$ day-ahead unit-commitment levels, $u_{\theta,\xi}^U$, $u_{\theta,\xi}^D$ reserve activation levels and $S_{\theta,\omega}^F$ and $S_{\theta,\omega}^X$ the sales of forwards and call options with strike price X ,

where the vector y represents the decisions of a representative retailer in the market

$$y = (B_\omega^F, B_\omega^X, p_\omega, e_\xi^X, \hat{p}_\xi, A_\omega^{NR}, A_\xi^R, \alpha_\xi^{NR}, \alpha_\xi^R)$$

with B_ω^F and B_ω^X the contracts procured in the day-ahead markets, p_ω the retail price for non-responsive consumers, $\hat{p}_\xi$ the real price for responsive consumers, A_ω^{NR} and A_ξ^R the fixed charges for non-responsive and responsive consumers, and α_ξ^{NR} and α_ξ^R the rationing of consumers in state ξ .

and the vector e represents the retailer's activation decisions $e_\xi^X \in [0, 1]$ of the call option with strike price X in state ξ .

(1) Generators maximize profit. A firm with technology $\theta \in \Theta$ maximizes expected profit given prices z^* and option activation levels e^*

$$x_\theta^* = \arg \max_{x_\theta} \mathbb{E}_\omega \mathbb{E}_{\xi|\omega} [\Pi_{\theta,\omega,\xi}(x_\theta; z^*, e^*)],$$

by choosing x_θ subject to dispatch constraints, the physical production constraint and the obligation for production to fulfill contract obligations:

$$R_{\theta,\omega}^D \leq Q_{\theta,\omega} \leq K_\theta - R_{\theta,\omega}^U, \quad (31)$$

$$q_{\theta,\xi} = Q_{\theta,\omega} + u_{\theta,\xi}^U R_{\theta,\omega}^U - u_{\theta,\xi}^D R_{\theta,\omega}^D, \quad u_{\theta,\xi}^U, u_{\theta,\xi}^D \in [0, 1], \quad (32)$$

$$q_{\theta,\xi} = S_{\theta,\omega}^F + \sum_{X \in \mathcal{X}} e_\xi^{*X} S_{\theta,\omega}^X. \quad (33)$$

(2) Retailers maximize profit. Retailers maximize expected profit by setting y and e subject to the consumer's participation constraints of the two consumer groups and the physical and contractual balancing constraints:

$$U_\omega^{NR} \leq \mathbb{E}_{\xi|\omega} [S_\xi(p_\omega, \alpha_\xi^{NR}) - p_\omega D_\xi(p_\omega, \alpha_\xi^{NR}) - A_\omega^{NR}], \quad (34)$$

$$U_\xi^R \leq \hat{S}_\xi(\hat{p}_\xi, \alpha_\xi^R) - \hat{p}_\xi \hat{D}_\xi(\hat{p}_\xi, \alpha_\xi^R) - A_\xi^R, \quad (35)$$

$$q_\xi^{\text{retail}} = \sigma D_\xi(p_\omega, \alpha_\xi^{NR}) + (1 - \sigma) \hat{D}_\xi(\hat{p}_\xi, \alpha_\xi^R), \quad (36)$$

$$q_\xi^{\text{retail}} = B_\omega^F + \sum_{X \in \mathcal{X}} e_\xi^X B_\omega^X. \quad (37)$$

(3) Markets clear. There is a market equilibrium in the day-ahead market for the forward and all option markets, and the total level of physical production equals

physical consumption

$$\sum_{\theta} S_{\theta,\omega}^F = B_{\omega}^F, \quad \sum_{\theta} S_{\theta,\omega}^X = B_{\omega}^X \quad (38)$$

$$\sum_{\theta} q_{\theta,\xi} = q_{\xi}^{\text{retail}} \quad (39)$$

(4) Free entry and consistency *The representative retailers make zero profit and the reservation utilities of consumers correspond to the utility they can obtain in equilibrium.*

We can now derive the following proposition:

Proposition 3. *The reserve market equilibrium obtains the first best outcome. The retailer will exercise call options iff their strike price X is less than the real-time retail price $\hat{p}_{\xi}$,*

$$e_{\xi}^X = \mathbf{1}_{\{X < \hat{p}_{\xi}\}} \quad \text{with } e_{\xi}^X \in [0, 1] \text{ if } X = \hat{p}_{\xi}.$$

The day ahead forward price and option price are given by

$$\begin{aligned} p_{\omega}^F &= \mathbb{E}_{\xi|\omega}(\hat{p}_{\xi}), \\ p_{\omega}^X &= \mathbb{E}_{\xi|\omega}\{\max\{\hat{p}_{\xi} - X, 0\}\}. \end{aligned}$$

which follows from the retailer arbitrage conditions of buying call options and balancing demand and supply with price instruments and rationing. Generators will self-select in selling contracts that correspond to their physical positions and the marginal adjustment costs of their technology:

$$\begin{aligned} S_{\theta,\omega}^F &= Q_{\theta,\omega}, \\ S_{\theta,\omega}^X &= R_{\theta}^U \mathbf{1}_{\{X = c_{\theta}^U\}} + R_{\theta}^D \mathbf{1}_{\{X = c_{\theta}^D\}}. \end{aligned}$$

The generator's dispatch maximizes profits by choosing between not producing, committing to a fixed quantity, selling upward reserves, or committing to a fixed quantity and downward reserves.

$$\lambda_{\theta,\omega} = \max\{0, p_{\omega}^F - c_{\theta}, p_{\omega}^{c_{\theta}^U} - I_{\theta}^U, p_{\omega}^{c_{\theta}^D} - I_{\theta}^D - c_{\theta} + c_{\theta}^D\}$$

Free entry will drive down the investment decisions of all technologies so:

$$\mathbb{E}[\lambda_{\theta,\omega}] = I_{\theta}$$

Chao and Wilson (1987) show that in a retail market, the social optimum can be implemented equivalently through spot price or an array of incentive-compatible contingent contracts from which consumers self-select their own contract by their willingness to pay, which is privately known. Also, Oren (2003) claims the equivalence of spot pricing and technology-specific energy and capacity payment to recoup investment costs. This paper, by contrast, demonstrates that a real-time market can be replaced by a forward market, combined with a reserves market with a menu of call options. The reserve market provides a price signal for flexibility in the absence of a real-time market, and we prove in the Appendix D that if there is only a forward market, the flexibility premium vanishes, leading to distorted investment in flexible technologies.

Example We provide an example to clarify the inefficiency of a forward-only market and how the reserve market compensates for the flexibility premium. There is one state in second stage ($|\Omega| = 1$) with two possible states: low demand: $\varepsilon = L$ with probability f_L , and high demand: $\varepsilon = H$ with probability f_H , $f_H + f_L = 1$; two technologies are available: $\theta \in \{1, 2\}$. Both technologies have the same production cost, $c_1 = c_2 = c$. Technology 1 is totally flexible, so it can postpone production until the state of the world is realized; technology 2 is totally inflexible, so it must commit itself to production before demand is known and is not able to adjust in real-time. That is, $I_1^U = I_1^D = 0$, $c_1^U = c_1^D = c_1$, and $I_2^U = I_2^D = c_2^U = c_2^D = \infty$.¹² The investment cost of technology 1 is higher than that of technology 2: $I_1 > I_2$. All consumers are responsive to real-time prices ($\sigma = 0$). The total capacity of each technology is denoted by K_1 and K_2 .

Denote real-time prices for both states as p_L and p_H , respectively. The social planner's objective is to maximize social welfare by choosing capacity K_1, K_2 , quantity committed by technology 2 before demand is realized Q_2 , and quantity produced by technology 1 in low state $q_{1,L}$ and in high state $q_{1,H}$:

$$\begin{aligned} \max_{\{K_1, K_2, q_{1,\varepsilon}, Q_2\}} & \mathbb{E}[S_\varepsilon(q_{1,\varepsilon} + Q_2) - c \cdot (q_{1,\varepsilon} + Q_2)] - I_1 K_1 - I_2 K_2. \\ \text{s.t.} \quad & q_{1,\varepsilon} \leq K_1, \\ & q_{2,\varepsilon} = Q_2, \\ & Q_2 \leq K_2. \end{aligned} \tag{40}$$

¹²By abuse of notation, we equalize the infinite numbers.

The first-order conditions yield:

$$\begin{aligned}
p_H &= \frac{I_1}{f_H} + c, \\
p_L &= c - \frac{I_1 - I_2}{f_L} < c, \\
Q_2 &= K_2; q_1^H = K_1; q_1^L = 0, \\
K_2 &= S_L^{-1'}(p_L), \\
K_1 &= S_H^{-1'}(p_H) - K_2,
\end{aligned}$$

Hence, in the presence of demand uncertainty and an efficient real-time market:

- a) inflexible firms earn the expected price $E(p) = c + I_2$;
- b) flexible firms earn p_H in high demand state and do not produce in low demand state;
- c) low demand price is below marginal production cost, $p_L < c$.

Flexible firms earn an expected premium equal to $I_1 - I_2$, which recoups the extra capacity investment cost of flexible assets. Now, consider the absence of a real-time market. Both types of firms need to determine the quantity and prices before demand is realized. The maximization problem becomes

$$\begin{aligned}
\max_{\{K_1, K_2, Q_1, Q_2\}} & E[S_\varepsilon(Q_1 + Q_2) - c \cdot (Q_1 + Q_2)] - I_1 K_1 - I_2 K_2. \\
s.t. \quad & Q_1 \leq K_1 \\
& Q_2 \leq K_2.
\end{aligned} \tag{41}$$

which gives

$$\begin{aligned}
K_1 &= 0; Q_2 = K_2 > S_L^{-1'}(p_L), \\
\mathbb{E}[S'(K_2)] &= c + I_2, \\
p^F &= E_\varepsilon(p), \\
p_L^R &= D_L^{-1}(K_2), \\
p_H^R &= D_H^{-1}(K_2).
\end{aligned}$$

p^F is the unit payment to generators, and p_H^R, p_L^R are the retail price charged to consumers in high and low states, respectively. As predicted, in the absence of

real-time markets and the presence of a forward market, long-term equilibrium shows:

- a) no investment in flexible technology;
- b) over-investment in inflexible technology;

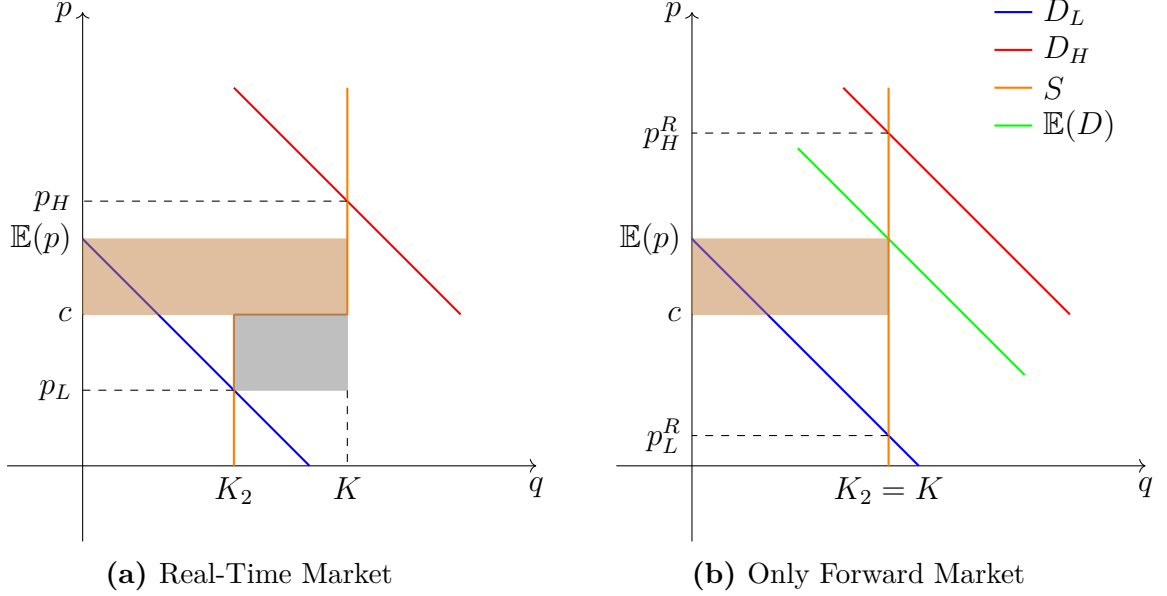


Figure 3: Equilibrium in Real-time and Forward Market

Note: This figure represents the equilibrium comparison between a real-time market and a forward only market. Sub-figure (3a) shows the equilibrium prices, quantities, and investments when a well-functioning real-time market exists. In this case, firms making a production commitment earn the expected price $\mathbb{E}(p)$ and the net profit is shaded in brown. In addition, flexible firms can react to real-time demand shocks so they avoid the loss at the low state (shaded in gray). Flexibility premium is unconditional loss avoidance, so the brown area times the probability of being in a low state p_L , which is equal to $I_1 - I_2$. Sub-figure (3b) shows the equilibrium when a real-time market is not available. Commitment is required for both flexible and inflexible firms, according to expected demand, and flexible firms are not rewarded for providing flexibility. Hence, in the long term, no flexible assets survive. Furthermore, to balance the welfare loss from under-consumption in the low state and rationing in the high state. $K_2(b) > K_2(a)$.

In this case, there is a missing market problem; firms and consumers cannot directly trade on goods as $p_R < c$; they will not produce even though there is a fixed payment. Hence, there must be an intermediary (e.g., retailer) that pays p^F to the firm and charges a two-part tariff from consumers.

Next, we show how an options market complements the forward market and achieves an optimum even without a real-time market. Firms and consumers can trade in either a forward market or an options market, or both. Firms receive p^F for the unit quantity they sell in the forward market; if they trade in the options

market, they receive the option price p^X for being available in real-time and X is the strike price. There is a huge penalty for not being available when activated to guarantee that only flexible firms enter the options market, and they will not sell more than they invest. The decision made by inflexible firms is trivial. They will produce up to capacity as long as the forward price p^F is higher than the production cost c . The objective function of flexible firms is

$$\begin{aligned} \max_{Q_1^F, Q_1^O, K_1} & (p^F - c)Q_1^F + p^X Q_1^O + \mathbb{E}[(X - c)q_{1,\varepsilon}] - I_1 K_1. \\ \text{s.t.} \quad & q_{1,\varepsilon} \leq Q_1^O, \\ & Q_1^F + Q_1^O \leq K_1. \end{aligned}$$

Q_1^F is quantity sold in the forward market; Q_1^O is quantity sold in the options market; $q_{1,\varepsilon}$ is activated options in state ε . Social optimum can be attained through a forward market with forward price $p^F = E_\varepsilon(p)$ and an options market with option price p^X and strike price X described by:

$$p^X = f_H(p_H - X), \quad p_L \leq X \leq p_H. \quad (42)$$

As shown in Lemma 7, when there is only one technology providing flexibility, there are infinitely many combinations of strike and activation prices. If the strike price is equal to the production cost, $X = c$, a flexible firm should be paid a capacity payment p^X larger than the opportunity cost of not trading in the forward market:

$$p^X > p^F - c. \quad (43)$$

Proof. When $X = c$, $p^X = f_H(p_H - c) = I_1 > I_2 = p^F - c$. $\square$

3.3 Existing reserve market designs

The design of reserve markets has evolved over time to address asymmetric information and avoid excessive payments to reserve providers. Nowadays, there are three main types of reserve markets. The first is an integrated market for energy and reserves, commonly used in the U.S. In Europe, a separate reserve market is used. There are two pricing schemes for reimbursing reserves: pay-as-bid and pay-as-clear. Below, we discuss these three market structures and whether they can efficiently reward the flexibility providers.

We have shown that the flexibility premium is not compensation for commitment or adjustment costs, but an option value that arises from the ability

to adjust output on short notice.¹³ Therefore, to minimize the details, we take $c_\theta^{U/D} = 0, I_\theta^U = 0$.

3.3.1 Integrated market for energy and reserves

Co-procurement of energy and reserves is widely used in U.S. electricity markets, such as ISO New England (ISO-NE) and the Electric Reliability Council of Texas (ERCOT). All generators submit their bids b_θ^X for energy or reserves, and the system operator clears the market with a uniform price p_ω^F . Based on the idea that providing reserves incur opportunity cost of not providing energy, generators that produce electricity receive p_ω^F , and the generators providing unemployed reserves receive the difference between the forward price and their own bid $p_\omega^F - b_\theta^X$ as capacity payment, and the bid b_θ^X serves as the strike price. If the cost information is perfectly known (Hogan, 2013; Oren, 2003), the capacity payment is $p_\omega^F - c_\theta$, but if the cost is private information, the generators would bid according to the distribution of the forward clearing price F_ω^F to maximize their expected profit:

$$\pi_{\theta,\omega} = \int_{b_\theta^X}^{\infty} (p_\omega^F - b_\theta^X) dF_\omega^F + (b_\theta^X - c_\theta)(1 - F_\omega(b_\theta^X))(1 - F_\omega^F(b_\theta^X)). \quad (44)$$

The optimal bidding is given by:

$$b_\theta^o = c_\theta - \frac{[1 - F_\omega^F(b_\theta^o)]F_\omega(b_\theta^o)}{F_\omega(b_\theta^o)[1 - F_\omega^F(b_\theta^o)] + F_\omega^F(b_\theta^o)[1 - F_\omega(b_\theta^o)]}. \quad (45)$$

Therefore, we obtain an underbidding result consistent with Oren and Sioshans (2005) that $b_\theta^X \leq c_\theta$, and the inequality strictly holds when reserve is provided. Recall that the option price is given by:

$$\pi_{\theta,\omega}^e = \int_{c_\theta}^{\infty} (p_\xi - c_\theta) dF_\omega. \quad (46)$$

Hence, efficiency is achieved when $b_\theta^X = c_\theta$ and $F_\omega^F(p_\xi) = F_\omega(p_\xi)$, which means there is no reserve requirement. We can further show that the efficient bidding for reserves is:

$$b_\theta^e = c_\theta - \frac{\int_{-\infty}^{c_\theta} (c_\theta - p_\xi) dF_\omega}{F_\omega(b_\theta^e)} \neq b_\theta^o. \quad (47)$$

Since opportunity cost is not a proper benchmark for reimbursing reserve provision, truthful bidding would lead to a distortion in flexibility investment. Although

¹³Although it is clear that a technology's flexibility premium decreases with the commitment and adjustment costs.

underbidding allows reserve providers to earn information rents, these rents are not equivalent to the flexibility premium in general. I summarize the result as follows:

Lemma 5. *In an integrated market, energy suppliers bid their true cost, while firms that provide reserves have an incentive to shade their bids. In the long run, the investment in flexibility is not efficient.*

3.3.2 Separate reserves market: pay-as-bid

We now discuss a separate reserve market that is prevalent in Europe. In practice, reserves are categorized by response time and the duration of activation: Primary reserves (Frequency Containment Reserves) act within 10 seconds, Secondary reserves (Automatic Frequency Restoration) are activated within 30 seconds and may last 5 to 15 minutes and Tertiary reserves (Manual Frequency Restoration) are dispatched within 5 to 15 minutes and may last up to a few hours.

Primary reserves are paid a pay-as-clear capacity payment, with no energy payment, which is similar to paying reserve providers a forward price, which would distort flexibility investment. Secondary reserves are paid a pay-as-bid capacity payment and a pay-as-bid energy payment. Tertiary reserves are paid on a pay-as-clear basis for energy and on a pay-as-bid basis for capacity. Both secondary and tertiary reserves are procured sequentially, with the capacity market clearing first. The selected generators then submit energy bids.

In the second stage, auction theory shows that a pay-as-bid rule leads to overbidding, and a pay-as-clear rule induces truthful reporting. We focus on the first stage. Denote b^{p^x} as the capacity bid for reserves. Since there is no commitment cost, the capacity cost consists only of the foregone profit in the forward energy market, $p_\omega^F - c$. $\mathbb{P}$ is the probability that a generator is selected to provide reserves. Let $\pi_{\theta,\omega}^E$ be the expected profit from reserve activation. The objective of a risk-neutral bidder indexed by θ is to maximize

$$\mathbb{E}[\pi(b_{\theta,\omega}^{p^x}, b_{\theta,\omega}^X)] = [b_{\theta,\omega}^{p^x} - (p_\omega^F - c_\theta) + \pi_{\theta,\omega}^E] \mathbb{P}_\omega(b_{\theta,\omega}^{p^x}, b_{-\theta,\omega}^{p^x}). \quad (48)$$

The optimal capacity bidding is expressed by

$$b_{\theta,\omega}^{p^x,o} = -\frac{1}{\ln \mathbb{P}_\omega(b_{\theta,\omega}^{p^x,o})'} + p_\omega^F - c_\theta - \pi_{\theta,\omega}^E. \quad (49)$$

where

$$\ln \mathbb{P}_\omega(b_{\theta,\omega}^{p^X,o}) = \frac{\partial \mathbb{P}_\omega(b_{\theta,\omega}^{p^X,o}) / \partial b_{\theta,\omega}^{p^X,o}}{\mathbb{P}_\omega(b_{\theta,\omega}^{p^X,o})}. \quad (50)$$

Since $\frac{\partial \mathbb{P}}{\partial b^{p^X}} \leq 0$ and $\mathbb{P} \geq 0$, it is straightforward to show that $b_{\theta,\omega}^{p^X,o} \geq p_\omega^F - c_\theta - \pi_{\theta,\omega}^E$. The intuition is that the capacity bid only affects the probability of being selected in the capacity auction, but will not influence the *order of merit* in activation. Therefore, reserve providers can earn information rent by bidding above the capacity cost. To equalize information rent and flexibility premium, the probability of being selected $\mathbb{P}_\omega$ is given by

$$\mathbb{P}_\omega(b^{p^X}) = e^{-\int_0^{b^{p^X}} \frac{1}{V_\omega(t)} dt}. \quad (51)$$

where $\nu_{\theta,\omega} = V_\omega(b_{\theta,\omega}^{p^X})$. Hence, the flexibility premium is a function of the capacity bid, which rules out the possibility that flexible firms with different flexibility premiums have the same equilibrium score. However, imposing that the information rent equals the flexibility premium yields $\frac{db^{p^X}}{d\theta} = 0$, implying that all types have the same bid in equilibrium. This contradicts the existence of a technology-specific flexibility premium. Therefore, for each state ω , the selection probability $\mathbb{P}$ can be designed such that at most one type of reserve provider can earn the efficient rent.

Lemma 6. *In a sequential reserve auction where the capacity market follows the pay-as-bid rule, each bidder bids higher than the difference between the foregone profit in the forward energy market and the expected profit from energy activation, which is given by Eq.(49). The energy bid is $b_\theta^X = c_\theta$ if energy price is set pay-as-clear and $b_\theta^X > c_\theta$ if energy price is set pay-as-bid. In the long run, flexibility investment is efficient if there is only one type of flexible asset.*

3.3.3 Separate reserves market: pay-as-clear

Finally, we consider the case in which both energy and capacity are paid on a pay-as-clear basis. This rule is not used in practice but was proposed as an incentive-comparative mechanism in a seminar paper by Chao and Wilson (2002). They argue that the energy payment should be cleared with a real-time price, and the capacity bid should be the difference between the foregone profit in the forward energy price and the expected profit from the reserve activation. Chao and Wilson (2002) find that the optimal capacity bid is negative with an example. Our paper shows that a negative capacity bid is not a special case but an inevitable result

of underestimating the profit that should be earned from providing flexibility. In theory, if energy provision is cleared at the real-time price, the capacity bid should be zero in the absence of risk aversion and price caps, implying that there is no need for a separate capacity market for reserves.

When both capacity and strike price are predetermined, pay-as-clear for reserves cannot achieve the long-term optimum, even if reserve suppliers recognize that they should earn the flexibility premium. For technology indexed by θ , the efficient expected profit in state ω should be:

$$\pi_{\theta,\omega} = p_{\omega}^F - c_{\theta} + \nu_{\theta,\omega}.$$

We propose two separate auctions for upward and downward reserves. As before, we use $X^{\{U,D\}}$ and $p^{X,\{U,D\}}$ to denote the strike price and option price for upward and downward reserves, respectively. If the equilibrium pay-as-clear price pair $(p^{X,\{U,D\}}, X^{\{U,D\}})$ exists, it should satisfy

$$p_{\omega}^{X,D} + [1 - F_{\omega}(c_{\theta}^U)][X^U - c_{\theta}] = \pi_{\theta,\omega}, \quad \forall \theta. \quad (52)$$

for upward reserves, and

$$p_{\omega}^{X,U} + F_{\omega}(c_{\theta})[X^D + c_{\theta}] = \nu_{\theta,\omega}, \quad \forall \theta. \quad (53)$$

for downward reserves. Hence, the system of linear equations in (52) or (53) has a unique set of solutions if and only if two technologies providing reserves, and these two constraints are consistent. There is no solution when more than two types of reserve providers are available, and multiple equilibria exist if there is only one flexible technology.

Nevertheless, in a two-stage pay-as-clear auction, competition among firms would imply truthful bidding, meaning they only consider foregone profit in the forward energy market and direct costs to provide reserves, if any, as a flexibility premium behaves as a reward instead of a cost. Since the quantity of required reserves is normally set inelastic, there is no scarcity rent. Therefore, the last technology that provides flexibility does not earn a flexibility premium if the system operator does not pay an extra payment, and a recursive inference shows that no flexible assets exist in the long run.

Lemma 7. *In a two-stage reserves auction with pay-as-clear, each bidder bids its true marginal cost $b_{\theta}^X = c_{\theta}$ in the second stage, and the difference of foregone profit in the forward energy market and expected energy profit $\pi_{\theta,\omega}^E$ as capacity bid*

$b_{\theta}^{p^x} = p_{\omega}^F - c_{\theta} - \pi_{\theta,\omega}^E$. However, flexibility investment is inefficient.

In short, a well-functioning real-time market is the only market-based approach to achieve efficient short-term pricing and long-term investment in flexibility. Reserve markets that centrally provide procurement menus can be a good compromise if real-time settlement is impossible. However, this market is demanding in design as it depends on the time-varying data specified by the system operator. It is, hence, highly sensitive to the information collected and its computation capability. Without extra reimbursement from the system operator, none of the market-based auctions for reserves can restore investment efficiency because they fail to price in technology-specific flexibility premiums. It is pivotal that each generator chooses among different levels of capacity payments in exchange for being available in real-time at the corresponding activation price. Technology-specific flexibility premium guarantees the selection is incentive-compatible. The comparison of different market designs is summarized below:

Proposition 4. *The social welfare of each market design is ordered such that:*

$$\begin{aligned} W(\text{real-time}) &= W(\text{reserve+forward}) \geq W(\text{pay-as-bid}) \\ &> W(\text{integrated}) = W(\text{pay-as-clear}) = W(\text{forward}) \end{aligned} \tag{54}$$

4 Conclusion

This article analyzes efficient pricing and investment of (in)flexible technologies under demand uncertainty. There are three takeaways. First, flexible assets should earn a flexibility premium on top of the expected price, and this can be efficiently reflected in a state-contingent real-time price. Second, a reserves market with a menu of call options, combined with a forward market, can also achieve the social optimum. Finally, current reserve market designs, which are primarily based on the opportunity costs of foregone energy production, are inefficient.

These results are relevant to electricity markets. It highlights the importance of a real-time or balancing market in providing proper incentives for flexibility investment. It also points to the limitations of the main proposed reserve market designs. In general, both integrated and separate reserve auctions lead to inefficient investment, as neither reimburses the technology-specific flexibility premium. In many European countries, both a balancing energy market and a capacity market have been developed, based on the argument that flexibility providers should be compensated for their commitment and adjustment costs through the capacity market. This view is not supported by our results. We show that balancing

prices can reimburse both the direct and indirect costs of adjustment. Therefore, a balancing capacity market is not necessary to complement the balancing energy market.

To isolate the flexibility premium arising from uncertainty, this paper abstracts from market failures such as risk aversion, market power, and externalities. The presence of these market failures can justify a reserve capacity market even when a well-functioning real-time market exists, but we argue that reimbursing direct adjustment costs is not a sufficient reason. With market failures, the decentralized market cannot achieve efficiency, and the premium a flexible firm can earn would also change. For instance, if flexible firms have market power, they may earn a higher flexibility premium. If firms are risk-averse, the value of flexibility would be lower, and a reserve market that reduces profit uncertainty could help restore efficiency. Preventing a blackout is a public good (Joskow & Tirole, 2007), and securing sufficient reserves in advance would increase the value of flexibility. However, these extensions do not change the main implication: flexibility providers should be compensated with a flexibility premium.

Moreover, this paper builds an asymmetric structure of inflexibility cost and introduces investment cost in flexibility. This complexity is not unnecessary, but observed from practice. One example is a wind farm, which is easy to turn off but not to turn on: its generation also depends on the weather, which cannot be controlled. This assumption motivates the separate purchase of incremental and decremental reserves. Furthermore, the investment cost in flexibility is also reasonable, as plants need a warm-up to produce. By this assumption, we not only have a rough idea about adjustment costs, but also how they explicitly enter the cost function and equilibrium. This cost structure is also welcomed to be tested by empirical research.

Last, even though this paper is in the context of the electricity market, a flexibility premium widely exists in other sectors, such as transportation and hotels: consumers choose to book non-refundable tickets or hotels or a refundable one with a flexible premium, or buy a ticket at the last minute with a more volatile price. Airlines provide flexibility by purchasing different sizes of airplanes and hiring more people as reserves. Our conclusion can also be applied to these fields.

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Appendix

A Proof of Proposition 1 (d)

Proof. Take derivative w.r.t $Q_{\theta,\omega}$, $R_{\theta,\omega}^D$ and $R_{\theta,\omega}^U$:

$$[Q_{\theta,\omega}] : \mathbb{E}_{\xi|\omega}[p_{\xi}] - c_{\theta} - \lambda_{\theta,\omega} + \mu_{\theta,\omega} + \varphi_{\theta,\omega}^Q = 0 \quad (\text{A.1})$$

$$[R_{\theta,\omega}^D] : \mathbb{E}_{\xi|\omega}[u_{\xi,\theta}^D(c_{\theta}^D - p_{\xi})] - I_{\theta}^D - \mu_{\theta,\omega} + \varphi_{\theta,\omega}^D = 0 \quad (\text{A.2})$$

$$[R_{\theta,\omega}^U] : \mathbb{E}_{\xi|\omega}[u_{\xi,\theta}^U(p_{\xi} - c_{\theta}^U)] - I_{\theta}^U - \lambda_{\theta,\omega} + \varphi_{\theta,\omega}^U = 0 \quad (\text{A.3})$$

(A.2) and (A.3) can be rewritten as

$$\mathbb{E}_{\xi|\omega}[\max\{c_{\theta}^D - p_{\xi}, 0\}] - I_{\theta}^D - \mu_{\omega} + \varphi_{\theta,\omega}^D = 0 \quad (\text{A.4})$$

$$\mathbb{E}_{\xi|\omega}[\max\{p_{\xi} - c_{\theta}^U, 0\}] - I_{\theta}^U - \lambda_{\omega} + \varphi_{\theta,\omega}^U = 0 \quad (\text{A.5})$$

which correspond to a put option and a call option respectively. If a firm does not invest in downward flexibility, $dR_{\theta,\omega}^D < dQ_{\theta,\omega} \Leftrightarrow \mu_{\theta,\omega} = 0$, (A.1) is reduced to a forward:

$$\mathbb{E}_{\xi|\omega}[p_{\xi}] - c_{\theta} - \lambda_{\theta,\omega} = 0 \quad (\text{A.6})$$

If a firm invests in downward flexibility, both (A.1) and (A.4) hold, and to sum up these two equations gives

$$\begin{aligned} & \mathbb{E}_{\xi|\omega}[p_{\xi}] - c_{\theta} + \mathbb{E}_{\xi|\omega}[\max\{c_{\theta}^D - p_{\xi}, 0\}] - I_{\theta}^D - \lambda_{\theta,\omega} = 0 \\ \Rightarrow & \mathbb{E}_{\xi|\omega}[p_{\xi}] - c_{\theta} - \mathbb{E}_{\xi|\omega}[\min\{p_{\xi} - c_{\theta}^D, 0\}] - I_{\theta}^D - \lambda_{\theta,\omega} = 0 \\ \Rightarrow & \mathbb{E}_{\xi|\omega}[p_{\xi} - c_{\theta}^D] - c_{\theta} - \mathbb{E}_{\xi|\omega}[\min\{p_{\xi} - c_{\theta}^D, 0\}] + c_{\theta}^D - I_{\theta}^D - \lambda_{\theta,\omega} = 0 \\ \Rightarrow & \mathbb{E}_{\xi|\omega}[\max\{p_{\xi} - c_{\theta}^D, 0\}] - [c_{\theta} - c_{\theta}^D] - I_{\theta}^D - \lambda_{\theta,\omega} = 0 \end{aligned}$$

Therefore, a combination of a forward and a put option is equivalent to a call option. $\square$

B Proof of Lemma 2

Proof. Using the definition of the flexibility premium, we can consider two cases. In case the non-responsive technology has a positive value $\lambda_{\theta,\omega}^{\text{nr}} = \pi_{\theta,\omega}^Q > 0$, the flexibility premium reflects the fact that (1) committing to downward reserves on top of the forward contract might provide the extra value $\pi_{\theta,\omega}^D$, (2) replacing the forward commitment with upward reserves might be more profitable: $\pi_{\theta,\omega}^U > \pi_{\theta,\omega}^Q$. So in that case we have:

$$\nu_{\theta,\omega} = \max\{0, \pi_{\theta,\omega}^D, \pi_{\theta,\omega}^U - \pi_{\theta,\omega}^Q\} \text{ if } \pi_{\theta,\omega}^Q > 0.$$

In case selling technology has no value, $\lambda_{\theta,\omega}^{\text{nr}} = 0$ and $\pi_{\theta,\omega}^Q \leq 0$, it could still be profitable to (1) schedule upward reserves instead, or (2) to schedule energy forward only in combination with downward reserves.

$$\nu_{\theta,\omega} = \max\{0, \pi_{\theta,\omega}^U, \pi_{\theta,\omega}^Q + \pi_{\theta,\omega}^D\} \text{ if } \pi_{\theta,\omega}^Q < 0.$$

Above we have shown that $\pi_{\theta,\omega}^U$ and $\pi_{\theta,\omega}^D$ can be interpreted as call and put options. Using the call-put parity

$$\pi_{\omega}^c(X, A) - \pi_{\omega}^p(X, B) = \mathbb{E}_{\xi|\omega}[p_{\xi}] - X + B - A.$$

The two remaining terms $\pi_{\theta,\omega}^Q + \pi_{\theta,\omega}^D$ and $\pi_{\theta,\omega}^U - \pi_{\theta,\omega}^Q$ can also be written as options. For instance:

$$\pi_{\theta,\omega}^Q + \pi_{\theta,\omega}^D = \mathbb{E}_{\xi|\omega}(p_{\xi}) - c + \pi_{\omega}^p(c^D, I^D) = \pi_{\omega}^c(c^D, I^D + c - c^D).$$

□

C Proof of Proposition 3

Proof. The procurement and activation strategies of the retailers are straightforward, and we focus on the self-selection of the generators. For any θ choosing upward reserves, incentive compatibility requires that $\forall \theta, X$,

$$\mathbb{E}_{\xi|\omega}\{\max(\hat{p}_{\xi} - c_{\theta}^{C/D}, 0)\} \geq \mathbb{E}_{\xi|\omega}\{\max(\hat{p}_{\xi} - X, 0)\} + [1 - F_{\omega}(X)][X - c_{\theta}^{C/D}] \quad (\text{C.1})$$

This is equivalent to

$$\int_{c_\theta^{C/D}}^X \hat{p}_\xi dF_\omega \geq [F_\omega(X) - F_\omega(c_\theta^{C/D})]c_\theta^{C/D} \quad (\text{C.2})$$

which always holds. $\square$

D Forward market without real-time market

We consider an environment in which all wholesale electricity is traded in a forward market before uncertainty is resolved. In contrast to the decentralized case in Section 3.1, there is no wholesale real-time market: generators and retailers contract a fixed quantity at a single forward price p_ω^F , determined prior to the realization of demand. Once the state of demand ξ is known, contracted positions and production levels are fixed, and there is no further wholesale adjustment or re-pricing.

To isolate the effect of replacing the wholesale real-time market with a wholesale forward market, we retain very efficient retailers that balance demand with their contracted supply in each state of the world. These retailers use rich two-part contracts, which allow retail prices to reflect the social value of electricity while protecting them from opportunistic consumer switching in the absence of a short-term wholesale market price.

The retailer balances an internal energy constraint in each state of the world. Given its fixed forward position q_ω^R , total retail demand must equal this contracted supply in every realized state ξ . To achieve this balance, the retailer relies on two instruments: real-time pricing for responsive consumers and rationing for non-responsive consumers. Through these mechanisms, scarcity is efficiently allocated across consumer groups and states of the world, even though there is no wholesale market in which production or prices can adjust ex post. Two-part tariffs for both consumer groups ensure that these instruments can be applied efficiently while maintaining the retailer's zero-profit condition.

We will set up the model in a structure similar to the decentralized equilibrium in Section 3.1, but with some important changes:

- (i) *Wholesale market.* Wholesale trading occurs only at the forward stage ω , where quantities are contracted at a forward price p_ω^F . There is no wholesale market in the real-time states ξ , and hence no price that reflects the social value of electricity ex post.

- (ii) *Retail market.* Retailers maintain balance internally in each realized state ξ , based on their fixed wholesale positions q_ω^R . They adjust retail demand through real-time pricing for responsive consumers and rationing for non-responsive consumers.
- (iii) *Firms' decisions.* Generators choose capacity K_θ and forward commitments $Q_{\theta,\omega}$ based on expected conditions. Reserve and activation variables $(R_{\theta,\omega}^U, R_{\theta,\omega}^D, u_{\theta,\xi}^U, u_{\theta,\xi}^D)$ remain feasible but yield no value when the wholesale price is fixed; in equilibrium, they are set to zero.
- (iv) *Market clearing and prices.* Wholesale balance holds ex ante at the forward stage,

$$\sum_{\theta} Q_{\theta,\omega} = q_\omega^R,$$

while retail balance holds ex post within each retailer through its internal mechanisms $(\hat{p}_\xi, \alpha_\xi)$. In equilibrium, the expected internal retail price will be equal to the forward price,

$$p_\omega^F = \mathbb{E}_{\xi|\omega}[\hat{p}_\xi].$$

Implications. Allowing retail-side adjustment mitigates inefficiencies on the demand side (responsive consumers still see a real-time price signal $\hat{p}_\xi$), but flexible *generation* cannot monetize variation in system value because the wholesale price is fixed at p_ω^F . The flexibility premium for supply therefore vanishes, leading to distorted investment in flexible technologies even under an efficient retailer.¹⁴

Definition 4 (Forward Market Equilibrium). *Let x_θ be the vector of decision variables of a firm with technology θ : production levels $q_{\theta,\xi}$, unit commitments $Q_{\theta,\omega}$, $R_{\theta,\omega}^U$, $R_{\theta,\omega}^D$, investments K_θ ,*

$$x_\theta = (q_{\theta,\xi}, Q_{\theta,\omega}, R_{\theta,\omega}^U, R_{\theta,\omega}^D, K_\theta).$$

Let y be the vector of decision variables of a retailer selling to non-responsive consumers: a fixed fee A_ω , a retail price for non-responsive consumers p_ω , and the rationing rule $\hat{\alpha}_\xi$, and decision variables of a retailer selling to responsive

¹⁴A less efficient and arguably more realistic alternative is to assume that, in the absence of a wholesale real-time market, the system operator applies non-price rationing (e.g., random curtailment) and retailers neither compete in rationing rules nor deploy state-dependent retail prices. Our results below then provide an *upper bound* on performance; the less efficient variant would strengthen the investment distortions.

consumers: a fixed fee $\hat{A}_\xi$, a retail prices for responsive consumers $\hat{p}_\xi$, and the rationing rule α_ξ , and the quantity purchased in the forward market q_ω^R :

$$y = (\mathcal{A}_\omega, p_\omega, \alpha_\xi, \hat{\mathcal{A}}_\xi, \hat{p}_\xi, \hat{\alpha}_\xi, q_\omega^R)$$

Let z be the vector of price variables: the forward price p_ω^F , the reservation utility of non-responsive consumers U_ω and the reservation utility of responsive consumers $\hat{U}_\xi$:

$$z = (p_\omega^F, U_\omega, \hat{U}_\xi)$$

The vectors x_θ^*, y^*, z^* form a competitive equilibrium if

- for a given z^* , x_θ^* maximizes the profit of a generator with technology θ ,
- for a given z^* , y^* maximizes the retailers profit,
- in each state ξ all markets clear:

$$q_{\Theta, \xi}^* = q_{\Theta, \omega}^* = \sigma \mathcal{D}_\xi(p_\omega^*, \alpha_\xi^*) + (1 - \sigma) \hat{\mathcal{D}}_\xi(\hat{p}_\xi^*, \hat{\alpha}_\xi^*).$$

- retailers and generators make zero profit,
- the reservation utility of non-responsive consumers U_ω and of responsive consumers $\hat{U}_\xi$ is consistent with the equilibrium contracts.

Now, we build the maximization problem of a generator with technology θ as:

$$\begin{aligned} x_\theta^* \in \arg \max_{x_\theta} \mathbb{E}_\xi \left[p_\omega^F q_{\theta, \xi} - I_\theta K_\theta - c_\theta Q_\omega - (I_\theta^U + u_{\theta, \xi}^U c_\theta^U) R_{\theta, \omega}^U - (I_\theta^D - u_{\theta, \xi}^D c_\theta^D) R_{\theta, \omega}^D \right] \\ \text{s.t.} \quad K_\theta \geq 0, \\ \forall \omega, \quad R_{\theta, \omega}^D \leq Q_{\theta, \omega} \leq K_\theta - R_{\theta, \omega}^U \quad \text{and} \quad Q_{\theta, \omega}, R_{\theta, \omega}^U, R_{\theta, \omega}^D \geq 0 \\ \forall \xi, \quad q_{\theta, \xi} = Q_{\theta, \omega} + u_{\theta, \xi}^U R_{\theta, \omega}^U - u_{\theta, \xi}^D R_{\theta, \omega}^D \quad \text{and} \quad u_{\theta, \xi}^U, u_{\theta, \xi}^D \in [0, 1] \end{aligned} \tag{D.1}$$

and the retailer's profit maximization is given by:

$$\begin{aligned} y^* \in \arg \max_y \mathbb{E}_\xi \{ \sigma [A_\omega + p_\omega \mathcal{D}_\xi(p_\omega, \alpha_\xi)] + (1 - \sigma) [\hat{A}_\xi + \hat{p}_\xi \hat{\mathcal{D}}_\xi(\hat{p}_\xi, \hat{\alpha}_\xi)] - p_\omega^F q_\omega^R \} \\ \text{s.t.} \quad \forall \omega, \quad \mathbb{E}_{\xi|\omega} [\mathcal{S}_\xi(p_\omega, \alpha_\xi) - p_\omega \mathcal{D}_\xi(p_\omega, \alpha_\xi) - \mathcal{A}_\omega] \geq U_\omega \\ \forall \xi, \quad \mathbb{E}_\xi [\hat{\mathcal{S}}_\xi(\hat{p}_\xi, \alpha_\xi) - \hat{p}_\xi \hat{\mathcal{D}}_\xi(\hat{p}_\xi, \hat{\alpha}_\xi) - \hat{\mathcal{A}}_\xi] \geq \hat{U}_\xi \\ \forall \xi, \quad \sigma \mathcal{D}_\xi(p_\omega, \alpha_\xi) + (1 - \sigma) \hat{\mathcal{D}}_\xi(\hat{p}_\xi, \hat{\alpha}_\xi) = q_\omega^R \end{aligned} \tag{D.2}$$

Let κ_ξ be the Lagrange multiplier of the retailer's retail quantity constraint.

Proposition 5. *The first-order conditions representing the optimum of the only forward market are:*

- (a) *No rationing for responsive consumers and optimal rationing for non-responsive consumers is such that the VOLL is equal to the retail price charged to responsive consumers:*

$$\hat{\alpha}_\xi = 1. \quad (\text{D.3})$$

$$\forall \xi, \frac{\partial \mathcal{S}_\xi}{\partial \alpha_\xi} / \frac{\partial \mathcal{D}_\xi}{\alpha_\xi} = \hat{p}_\xi \quad \text{or} \quad \alpha_\xi \in \{0, 1\}. \quad (\text{D.4})$$

- (b) *The retail price p_ω for non-responsive consumers is:*

$$\forall \omega \quad \mathbb{E}_{\xi|\omega} \left[\frac{\partial \mathcal{S}_\xi}{\partial p_\omega} - p_\omega^F \frac{\partial \mathcal{D}_\xi}{\partial p_\omega} \right] = 0. \quad (\text{D.5})$$

- (c) *Forward prices equal the expected real-time retail price*

$$p_\omega^F = \mathbb{E}_{\xi|\omega}[\hat{p}_\xi].$$

- (d) *The zero-profit condition determines the fixed parts $\mathcal{A}_\omega$ and $\hat{\mathcal{A}}_\xi$:*

$$\mathcal{A}_\omega = (p_\omega^F - p_\omega) \mathbb{E}_{\xi|\omega}[\mathcal{D}_\xi(p_\omega, \alpha_\xi)],$$

$$\hat{\mathcal{A}}_\xi = (p_\omega^F - \hat{p}_\xi) \hat{\mathcal{D}}_\xi(\hat{p}_\xi, \hat{\alpha}_\xi).$$

- (e) *No commitment to reserves:*

$$R_{\theta,\omega}^U = 0, R_{\theta,\omega}^D = 0. \quad (\text{D.6})$$

- (f) *Energy commitment:*

$$p_\omega^F - c_\theta = \lambda_{\theta,\omega} - \varphi_{\theta,\omega}^Q. \quad (\text{D.7})$$

- (g) *Capacity investment:*

$$\begin{aligned} \mathbb{E}_\omega[\lambda_{\theta,\omega}] &= I_\theta \quad \text{if} \quad K_\theta > 0 \\ \text{Otherwise, } K_\theta &= 0 \end{aligned} \quad (\text{D.8})$$

Discussion In equilibrium, reserves are not used because, under our assumptions, keeping upward reserves available and activating them in all states would be strictly more costly than producing the contracted quantity directly. The value of a technology in state ω is therefore determined solely by its ability to sell a fixed production quantity in the forward market at the price p_ω^F . Formally, the unit return to capacity is given by

$$\lambda_{\theta,\omega}^F = \max\{p_\omega^F - c_\theta, 0\},$$

which is a similar payment that a completely inflexible technology would obtain in a real-time market. Hence, technologies with different degrees of operational flexibility receive identical expected revenues, and the flexibility premium that exists in the real-time market equilibrium disappears. Because forward prices are endogenously linked to investment choices, the resulting equilibrium supports a different long-run investment mix than in the decentralized real-time market.

Under random rationing, both high- and low-value non-responsive consumers are curtailed proportionally within each realized state ξ . Using $\mathcal{D}_\xi(p_\omega, \alpha_\xi) = \alpha_\xi \mathcal{D}(p_\omega)$ and the forward-market condition $\mathbb{E}_{\xi|\omega}[\hat{p}_\xi] = p_\omega^F$, the first-order condition for p_ω can be written as

$$p_\omega = p_\omega^F + \frac{\text{Cov}_{\xi|\omega}(\hat{p}_\xi, \alpha_\xi)}{\mathbb{E}_{\xi|\omega}[\alpha_\xi]}.$$

Compared to the decentralized case, the structure of this expression is similar, but the covariance term differs in its weighting. In the decentralized market, the corresponding term involves $\text{Cov}_{\xi|\omega}(p_\xi, \alpha_\xi \partial \mathcal{D}_\xi / \partial p_\omega)$, reflecting that price adjustments there are weighted by the marginal responsiveness of demand. Under proportional rationing, these slope terms cancel out, so the covariance in the forward-only market depends only on the co-movement between the internal retail price $\hat{p}_\xi$ and the rationing share α_ξ .

The average of the state-contingent transfers $\hat{A}_\xi$ does not, in general, equal zero. From $\hat{A}_\xi = (p_\omega^F - \hat{p}_\xi) \hat{\mathcal{D}}_\xi$ and $\mathbb{E}_{\xi|\omega}[\hat{p}_\xi] = p_\omega^F$, we obtain

$$\mathbb{E}_{\xi|\omega}[\hat{A}_\xi] = - \text{Cov}_{\xi|\omega}(\hat{p}_\xi, \hat{\mathcal{D}}_\xi).$$

Because flexible demand declines in high-price states, this covariance is negative, so the expected fixed payment $\mathbb{E}_{\xi|\omega}[\hat{A}_\xi]$ is positive. This positive transfer is not a reward for providing flexibility, but a financial adjustment that offsets the mis-

match between the fixed wholesale forward price and the varying retail revenues in different states of demand.